

VLSI Internship - Task 1

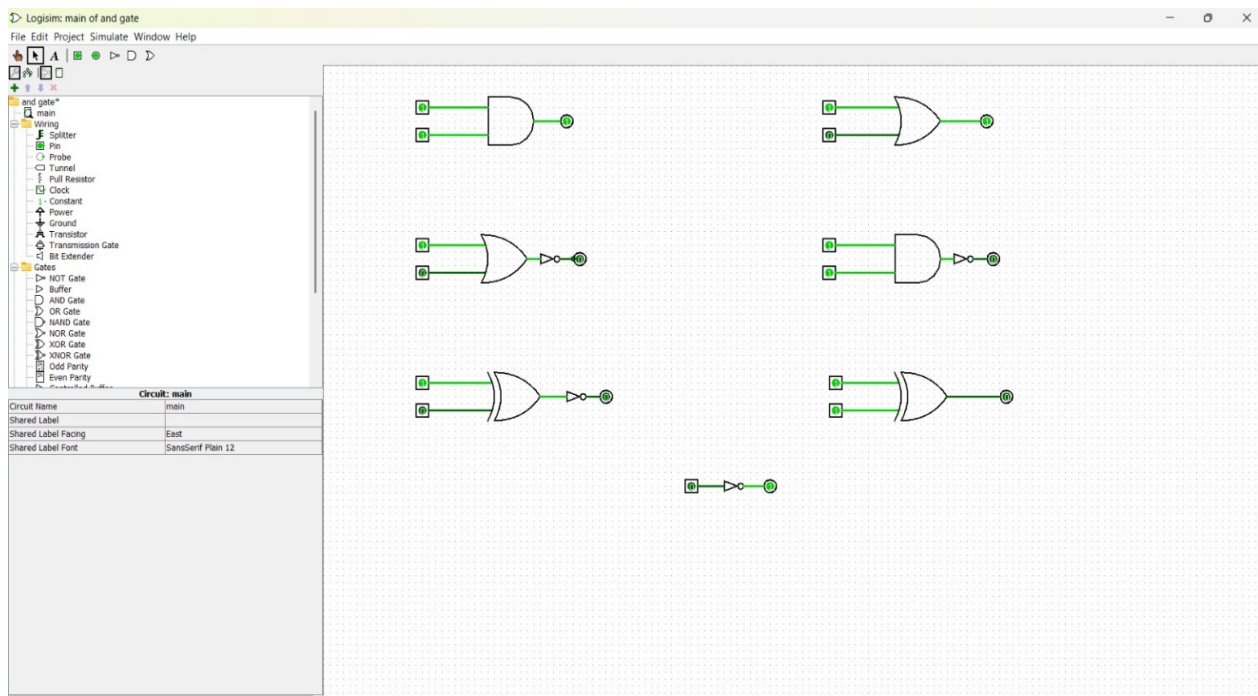
1. VLSI Design Flow

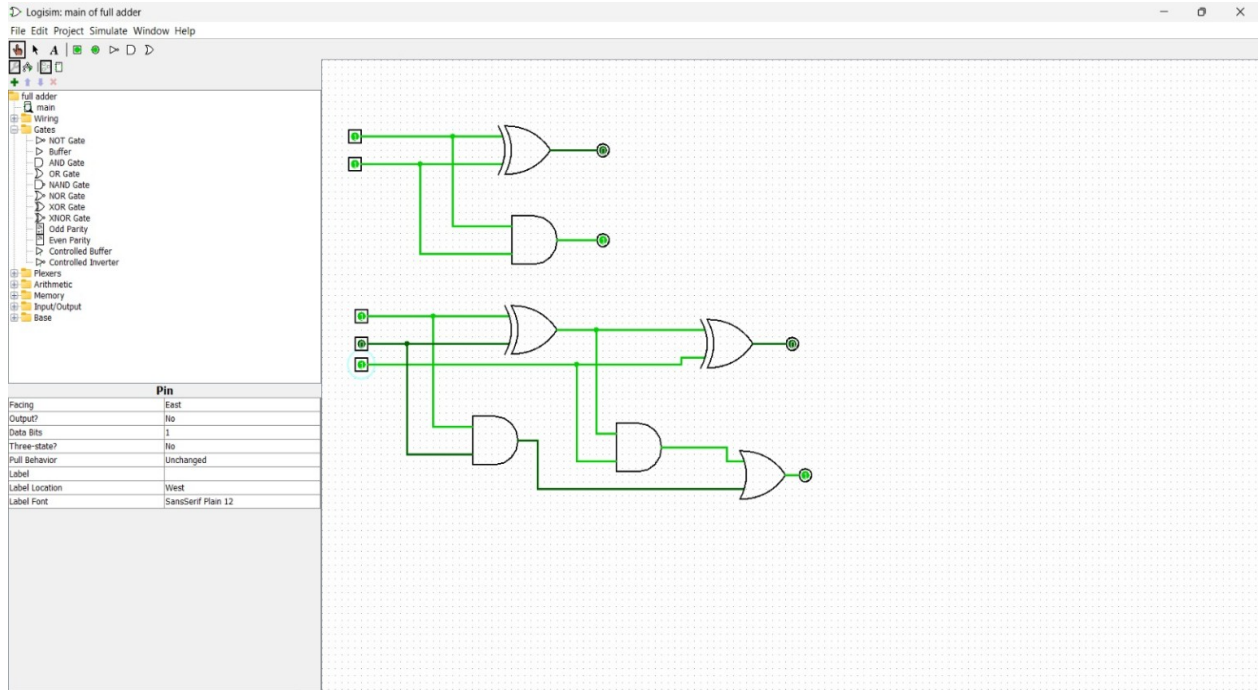
Specification → Architecture Design → RTL Design → Functional Verification → Synthesis → Physical Design → Timing & Power Analysis → Fabrication & Testing. The design requirements are first defined, then implemented in HDL, verified, converted into logic gates, physically laid out, analyzed for timing and power, manufactured and finally tested.

2. Truth Table of Logic Gates

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

3. Simulation Screenshots





4. Observations from Simulation

1. All basic logic gates produced outputs matching their truth tables.
2. The Half Adder correctly generated $\text{Sum} = A \text{ XOR } B$ and $\text{Carry} = A \text{ AND } B$.
3. The Full Adder correctly generated $\text{Sum} = A \text{ XOR } B \text{ XOR } \text{Cin}$ and $\text{Carry} = AB + AC_{\text{in}} + BC_{\text{in}}$.
4. The Logisim simulation confirmed correct operation without logical errors.